

Group 2: Predicting Game Recommendations from Steam User Reviews: A Multi-modal Sentiment and Users' Behavioral Analysis

Name: Bokyoung Son

Halicioğlu Data Science Institute (HDSI)

Email Address: boson@ucsd.edu

Name: Brandon Rocchio

Halicioğlu Data Science Institute (HDSI)

Email Address: brocchio@ucsd.edu

Name: Paige Saengsouvanna

Halicioğlu Data Science Institute (HDSI)

Email Address:

psaengsouvanna@ucsd.edu

Name: Hangyi Zhao

Halicioğlu Data Science Institute (HDSI)

Email Address: haz214@ucsd.edu

Abstract—User reviews on platforms such as Steam serve as essential indicators for game discovery; however, they vary considerably in sentiment, quality, and usefulness. This paper proposes a multi-modal machine learning framework that concurrently addresses three predictive tasks using the Steam Review Dataset (comprising over 100 million reviews): (1) recommendation classification—predicting whether a user recommends a game; (2) helpfulness prediction—assessing the usefulness of a review to other users; and (3) personalized game recommendation—ranking games that a user is likely to engage with next. We integrate review text features (TF-IDF and BERT embeddings) with behavioral signals (playtime, voting history, and engagement metrics) across gradient-boosted, collaborative filtering, and re-ranking models. LightGBM integrating both text and metadata achieves a ROC-AUC of 0.870 and increases the recall for the negative class from 50% to 66%. The BM25-weighted ALS method enhances Hit Rate@10 by 66% compared to a popularity baseline, whereas the Genre+Sentiment re-ranking approach attains the optimal balance between accuracy and diversity. Predicting helpfulness remains challenging (best $R^2 = 0.067$), indicating that platform-level factors beyond review content predominantly influence perceived usefulness.

Keywords — *Steam reviews, catalog coverage, recommendation systems, sentiment analysis, LightGBM, TF-IDF, Alternating Least Squares (ALS), helpfulness prediction, diversity analysis, BM25 weighting, genre entropy, collaborative filtering, BERT embeddings*

I. INTRODUCTION

Steam hosts over 100 million user reviews spanning more than 50,000 games, thereby establishing itself as one of the largest platforms for community-driven game evaluation. These reviews significantly influence purchasing decisions and inform recommendation algorithms, despite considerable variation in sentiment, depth, and perceived usefulness. Current recommendation systems on these platforms often encounter difficulties in simultaneously capturing textual sentiment, behavioral engagement signals, and review quality, as they frequently rely on a single modality or task in

isolation. This paper addresses this gap by conducting a comprehensive multi-modal analysis of Steam review behavior across three interrelated predictive tasks. The input to our system comprises a Steam user review accompanied by behavioral metadata, including playtime at the time of the review, voting history, and reviewer credentials. Our outputs comprise (1) a binary recommendation label indicating whether the reviewer endorses the game, (2) a continuous helpfulness score estimating the review's utility to the community, and (3) a ranked list of personalized game recommendations for each user. Our motivation is threefold in nature. Firstly, recommendation classification benefits from incorporating behavioral signals alongside textual data; for example, a user who has played for 200 hours before submitting a review offers different insights than one who has played for only 30 minutes. Second, the prediction of helpfulness holds practical significance for platforms that prioritize high-quality reviews; however, it remains insufficiently explored within gaming contexts. Third, personalized recommendation systems must balance accuracy with catalog diversity to address popularity bias, a well-established failure mode wherein models consistently recommend a limited subset of dominant titles. We demonstrate that the integration of textual and behavioral features consistently surpasses single-modality baselines across all three tasks, and that diversity-aware re-ranking can substantially improve catalog coverage without sacrificing accuracy.

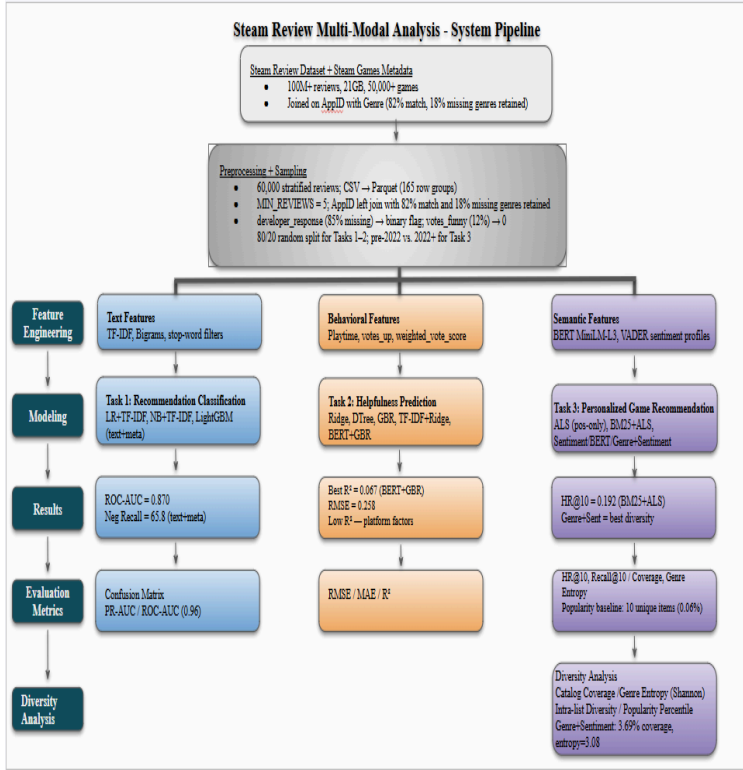


Fig 1. Project Modeling Pipeline

II. RELATED WORKS

A. Recommendation Classification

Traditional approaches to predict review sentiment typically rely on TF-IDF feature representations combined with linear classifiers that serve as interpretable baselines for text classification tasks. Lin et. al finds that Steam review data demonstrate that positive and negative gameplay experiences tend to reflect review sentiment [8]. Further recent research has explored contextual language models, such as BERT, which learn semantic relationships between words and can capture contextual meaning, sarcasm, and longer linguistic dependencies that simpler bag-of-words models may miss [2]. However, many review datasets exhibit substantial class imbalance, with positive recommendations dominating the negative ones. This imbalance can cause biased models predicting the majority class and reduce recall for negative reviews. Class weighting, threshold tuning, or alternative evaluation metrics are important considerations when training review-classification systems. Consequently, modern review prediction pipelines often combine strong interpretable baselines with more complex language models, while addressing dataset imbalance.

B. Review Helpfulness Prediction

Eberhard et al. study factors influencing the helpfulness of user reviews on Steam [22]. They analyze a dataset of over 132,000 reviews from 64 video games and derive 58 features related to review content, structure, readability, sentiment, and

user behavior. They apply statistical hypotheses and train a Random Forest classifier to predict review helpfulness categories. Wang et al. examine the relationship between review characteristics and helpfulness using regression analysis, and later convert the task into a binary classification problem based on helpful vote thresholds, using Random Forest and Gradient Boosting Decision Tree models for prediction [23]. Baowaly et al. study Steam review helpfulness using both classification and regression models on over 100,000 reviews from more than 4,400 games, incorporating metadata, lexical, semantic, and embedding-based features such as TF-IDF, LDA, and Word2Vec [24]. While these studies explore various contextual and textual features, most approaches rely heavily on hand-crafted features and threshold-based labels, rather than more recent end-to-end text representation methods.

C. Collaborative Filtering and Recommender System

Baltrušaitis et al. propose a comprehensive taxonomy of multimodal learning that advocates for the integration of textual and behavioral modalities in recommendation tasks, demonstrating that the fusion of heterogeneous signals consistently yields superior performance compared to single-modality approaches [1]. Sentiment-aware re-ranking improves collaborative filtering by aligning item-level sentiment profiles with user preference signals derived from review histories; notably, Hutto and Gilbert's VADER framework provides a lightweight, rule-based scoring mechanism that is particularly suited for this purpose [5]. Semantic embeddings based on BERT have also been employed as post-hoc re-ranking signals, wherein the cosine similarity between user and item embedding profiles—extracted from review texts—produces a content-aware relevance score that complements the foundational collaborative filtering recommendations [2].

D. Diversity and Popularity Bias

A well-documented failure mode of accuracy-optimized recommender systems is popularity bias, the tendency to recommend a small set of dominant items at the expense of catalog diversity. Hassan et al. Demonstrate that engagement metrics alone are inadequate for evaluating recommendation quality in platform settings, thereby motivating the incorporation of coverage and diversity metrics alongside hit rate [4]. McIlroy et al. similarly, demonstrate that surface-level engagement indicators can obscure significant variations in user preferences [9]. Shannon entropy over recommended genre distributions and catalog coverage percentage have emerged as practical diversity metrics for evaluating whether systems expose users to a breadth of content rather than concentrating recommendations on a narrow set of popular titles. These considerations directly inform our Genre+Sentiment re-ranking approach and the diversity analysis presented in this study.

III. DATASET AND FEATURES

A. Dataset

We use the Steam Review Dataset collected from the Kaggle platform [25,26]. The dataset contains over 100 million user reviews with rich metadata including review text,

playtime statistics, and community feedback signals. We also incorporate a game metadata dataset, merged with the reviews via `app_id`, which adds game-level attributes such as genre. Due to computational constraints, we sample a subset of reviews for model training and evaluation. Exploratory data analysis reveals several notable patterns in the dataset. First, the number of reviews increases rapidly after 2018, reflecting the growing popularity of the Steam platform. Second, users who recommend games generally exhibit higher playtime, suggesting that engagement signals can be informative behavioral features. Third, many structured features show weak linear correlations with outcome variables, indicating that nonlinear models may be needed to capture more complex relationships. Finally, popular games tend to receive significantly more reviews and recommendations, revealing a strong popularity bias in the platform.

Given the rich information contained in the dataset, several predictive tasks can be formulated. For example, the dataset can be used to predict whether a user recommends a game, estimate the perceived helpfulness of reviews, or build personalized game recommendation systems based on user behavior. In this project, we explore three complementary tasks: recommendation classification, review helpfulness prediction, and personalized game recommendation. These tasks capture different aspects of user-generated content on the Steam platform: recommendation classification reflects player satisfaction, helpfulness prediction measures community evaluation of review quality, and personalized recommendation models users' preferences.

B. Preprocessing & Features

The raw Steam review dataset was converted from CSV to Parquet format and partitioned into 165 row groups to facilitate memory-efficient streaming reads, thereby avoiding the necessity of loading the entire 21GB dataset into memory. A stratified sampling method was employed to select 60,000 representative reviews from the entire corpus. User eligibility was determined by imposing a minimum requirement of five reviews per user (`MIN_REVIEWS = 5`), resulting in 5,394,398 eligible users out of a total of 37,918,537 unique users.

For Tasks 1 and 2, an 80/20 random train-test split was implemented using a fixed random seed to ensure reproducibility. For Task 3, a temporal split was employed: reviews prior to 2022 constituted the training set, while those from 2022 onward comprised the test set, thereby preventing data leakage from future interactions into the training signal. This resulted in 177,894 training reviews and 53,942 test reviews from a sample of 20,000 users, averaging 11.6 reviews per user.

Missing values were managed as follows: the `developer_response` field, which was missing in 85% of cases, was converted into a binary indicator of presence rather than being imputed, as the absence of a response carries meaningful information. The `votes_funny` field, which has 12% missing values, was imputed with zero to indicate that newly posted reviews had not yet received any funny votes. Reviews containing fewer than 20 characters were excluded from text-based modeling pipelines due to insufficient information for reliable sentiment analysis or embedding. The

`steam_china_location` column contained no values throughout the dataset and was consequently removed. Unix timestamps were converted into datetime objects, and playtime values were transformed from minutes to hours to improve interpretability.

C. Feature Engineering

The features were categorized into three groups: behavioral, text-based, and semantic.

Behavioral features were directly extracted from the review metadata and encompass `playtime_hours` (converted from minutes), `votes_up`, `weighted_vote_score`, `has_developer_response` (a binary indicator), and `helpful_vote_ratio`. Game-level metadata features include `Game_Genres` and `popularity_tier`, the latter of which was calculated using quantile bins based on per-game review counts to represent relative prominence within the catalog.

For text-based features, TF-IDF vectorization was employed with a maximum vocabulary size of 40,000 tokens, a bigram range of (1,2), stop-word filtering, and a minimum document frequency threshold of five. This representation was used for recommendation classification (Task 1) and helpfulness prediction (Task 2).

Semantic features for Task 3 were developed using two distinct approaches. Initially, the VADER sentiment scoring method [5] was applied to each review text to generate a compound sentiment score. Sentiment profiles for each user and item were subsequently derived by averaging the scores across all training reviews, resulting in `avg_user_sentiment` and `avg_item_sentiment`. The resulting sentiment distribution across the items comprised 7,327 neutral, 5,322 positive, 506 negative, and 255 very positive instances, indicating an overall positive skew in the corpus. Second, dense embeddings were generated using the paraphrase-MiniLM-L3-v2 sentence transformer model, producing 384-dimensional vectors for each review. A minimum review length of 20 characters was enforced prior to embedding to exclude uninformative short-form text. A total of 126,886 reviews were embedded, yielding item-level profiles for 15,065 games and user-level profiles for 18,752 users, corresponding to a user coverage rate of 93.8%. For the remaining 6.2% of users with insufficient review history, BERT-based re-ranking was not applied.

Game genre information was extracted from the Steam Games metadata and joined on AppID. User genre preference profiles were developed from training history by aggregating genre counts of positively reviewed games, serving as the foundation for the Genre+Sentiment re-ranking approach in Task 3. Among the 60,000 sampled reviews, 18% lacked genre information following the AppID join; these reviews were retained for Tasks 1 and 2, but their genre features were regarded as absent in Task 3 for genre profiling.

IV. METHODS

A. Recommendation Classification

We model the task of predicting whether a Steam user recommends a game as a binary classification problem, where `voted_up` indicates whether a review is recommended with

target variable $y \in \{0,1\}$. Review text was first converted into numerical features using Term Frequency-Inverse Document Frequency (TF-IDF), which measures how important a word is within a specific review compared to the entire collection of reviews. For a word t in review d , the TF-IDF value is defined as $\text{TFIDF}(t,d) = \text{TF}(t,d) \times \log(N/\text{df}(t))$, where $\text{TF}(t,d)$ represents how frequently the word appears in the review, N is the total number of reviews, and $\text{df}(t)$ is the number of reviews containing that word.

We first trained two baseline models, Logistic Regression and Naive Bayes, widely used for text classification because they are simple, fast, and perform well on high-dimensional text features, such as TF-IDF vectors. Logistic Regression is a linear classification model used to estimate the probability that the user recommends the game at the time of the review. The model takes the TF-IDF vector x and computes a weighted sum $w^T x + b$, where w represents learned feature weights and b is a bias term. This value is passed through the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

which converts it into a probability between 0 and 1. The model predicts a recommendation when this probability exceeds the assigned threshold value, 0.5. During training, Logistic Regression learns weights that minimize cross-entropy loss, which measures the difference between the predicted probabilities and the true labels. Since each feature has a learned weight, the model is relatively interpretable and can reveal which words or features are associated with positive or negative recommendations.

Multinomial Naive Bayes is a probabilistic classification algorithm commonly used for text data. The model applies Bayes' theorem to estimate the probability that a review belongs to a particular class (recommended or not recommended) given the words it contains. The prediction is based on the formula: $P(y|x) = \frac{P(x|y)P(y)}{\sum_j P(x|j)P(j)}$ where $P(y)$ is the prior probability of a class and $P(x|y)$ represents the likelihood of observing the words in the review given that class. The model assumes that words appear independently given the class label, hence "naive", which simplifies the computation to $P(y) \prod_j P(x_j|y)$. Naive Bayes performs well because it effectively captures differences in word frequencies between positive and negative reviews.

We selected those models to balance interpretability, efficiency, and predictive performance. However, these linear models may struggle to capture more complex relationships between review text and metadata. To address this, we also deployed LightGBM, a gradient boosting decision tree model that can learn nonlinear interactions between features. Gradient boosting builds multiple decision trees sequentially, where each new tree focuses on correcting prediction errors made by earlier trees. This model optimizes the loss function, binary log loss, which measures how close predicted probabilities are to the true labels. In this project, LightGBM used both text features from TF-IDF and metadata features such as playtime, number of games owned, and review timing. By learning these interactions between these features, the model can capture patterns, like users with higher playtime being more likely to recommend a game or certain words having different meanings depending on the context. This

allows LightGBM to achieve stronger predictive performance than simpler baseline models.

B. Helpfulness Prediction

We treat review helpfulness prediction as a regression problem. For each review 'i', the model takes a feature vector ' x_i ' and predicts a helpfulness score ' y_i '. The target variable is the Steam 'weighted_vote_score', which represents how helpful the community perceives a review. The prediction task can be written as: $\hat{y}_i = f(x_i)$, where x_i contains the features extracted from review i and $f(\cdot)$ denotes the prediction model.

We use two types of features: 'structured metadata' and 'text features'. Structured features describe reviewer behavior and review context, including the number of reviews written by the author, number of games owned, total playtime, playtime at review time, and purchase-related flags such as 'steam_purchase', 'received_for_free', and 'early_access'. These features capture signals about reviewer experience and engagement.

To incorporate textual information from review content, we consider two approaches. First, we use 'TF-IDF' representations as a bag-of-words baseline: $\text{tfidf}(t,d) = \text{tf}(t,d) * \log(N / \text{df}(t))$, where N is the number of documents and $\text{df}(t)$ is the document frequency of term t . In addition, we use BERT embeddings to obtain dense semantic representations of review text.

For prediction models, we first evaluate simple baselines including Ridge Regression and Decision Tree Regression using structured features. To capture more complex nonlinear relationships, we employ Gradient Boosting Regression, which builds an ensemble of decision trees sequentially, where each tree is trained to reduce the residual error of the previous model.

C. Personalized Game Recommendation

All recommendation models were evaluated using a temporal train/test split, where interactions prior to 2022 formed the training set and interactions from 2022 onward formed the test set. The evaluation was performed on a sample of 20,000 users, each with a minimum of five reviews, using Hit Rate at $k=10$ (HR@10) and Recall@10 as the primary metrics.

Popularity Baseline: Each user was simply recommended the top 10 games with the most reviews from the dataset.

ALS (Positive-Only): The alternating least squares model factorizes the user-item interaction matrix into two low-dimensional matrices U and V representing user and item latent factors. Positive user interactions are measured via the voted_up column which is a binary indicator of if the user voted the game up. The model optimizes the following:

$$\min_{U,V} \sum_{u,i} c_{ui} (r_{ui} - u_u^T v_i)^2 + \lambda (||u_u||^2 + ||v_i||^2)$$

where r_{ui} is the observed interaction between user u and item i , c_{ui} is the confidence weight to the interaction, and λ is the regularization parameter. ALS solves this

objective by fixing one matrix iteratively while solving for the other.

BM25+ALS: Adding BM25 weighting to the ALS model rescales interaction counts to minimize the dominance of popular items and users with many reviews. For each interaction between a user U and item i the weighting is calculated with:

$$w_{ui} = \frac{tf_{ui}(k_1 + 1)}{tf_{ui} + k_1 \left(1 - b + b \frac{|u|}{avg_u}\right)}$$

where tf_{ui} is the interaction frequency, $|u|$ is the number of interactions for user u , and k_1 and b are hyperparameters for saturation and normalization.

Sentiment Boost: A re-ranking method was implemented on ALS candidates utilizing VADER sentiment scores [5]. An average sentiment profile ($avg_user_sentiment$) was calculated for each user based on their training reviews. The candidate items were subsequently re-ranked based on a composite score:

$$score(u, i) = ALS_score(u, i) + w \cdot (item_sentiment(i) - avg_user_sentiment(u))$$

where w is the boost weight. A sweep over $w \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ was conducted, with $w=0.5$ yielding the best $HR@10$. An increased voted-up ratio was also assessed as an alternative signal; however, it consistently impaired performance across all tested weights and was consequently omitted from the final system.

BERT Re-ranking: Dense semantic embeddings were generated for each review using the paraphrase-MiniLM-L3-v2 sentence transformer, yielding 384-dimensional vectors. Average embeddings were calculated for each user and each item based on their respective training reviews. Candidate items from ALS were re-ranked based on the cosine similarity between user and item embedding profiles:

$$score(u, i) = ALS_score(u, i) + w \cdot cosine(emb_u, emb_i)$$

where emb_u and emb_i are the mean BERT embeddings for user u and item i respectively. A sweep over $w \in \{0.3, 0.5, 0.7\}$ was conducted, with $w=0.3$ yielding the best $HR@10$. Users without sufficient review history for embedding (6.2% of users) received standard ALS scores without re-ranking.

Genre+Sentiment Re-ranking: A Genre+Sentiment re-ranking method was developed to optimize accuracy and catalog diversity simultaneously. A user genre preference profile was created by aggregating genre counts from all positively reviewed games in the training history. Candidate items were ranked by combining ALS scores with sentiment alignment and genre affinity to prioritize those matching the user's past preferences while keeping sentiment consistent. This method was developed to counter the popularity bias in the ALS baseline, where recommendations focused on a small set of dominant titles.

V. EXPERIMENTS, RESULTS, AND DISCUSSION

A. Recommendation Classification

To evaluate the task of predicting whether a Steam review recommends a game (voted `up`), we used a dataset of Steam reviews containing both review text and user metadata. To avoid bias from highly popular games dominating the dataset, we capped the number of reviews per game at 500 reviews per AppID, resulting in a dataset of 73,661 reviews across 862 games. The dataset is naturally imbalanced: approximately 79.8% of reviews are positive (recommended) and 20.2% are negative, reflecting the true distribution of Steam reviews.

We split the data into 80% training and 20% testing sets using stratified sampling to preserve the class distribution across splits. Review text was transformed into numerical features using TF-IDF vectorization, which balances vocabulary coverage and computational efficiency while filtering out rare words. We used unigrams and bigrams with a maximum vocabulary of 40,000 features and removed English stopwords to reduce noise. Logistic Regression was trained with L2 regularization and a maximum of 1000 iterations. The Multinomial Naive Bayes model used a smoothing parameter $\alpha=0.5$. For the LightGBM model, we used 2000 trees with class weights balanced to account for class imbalance.

To evaluate model performance, we used several standard classification metrics: accuracy, precision, recall, and F1 score. Since the dataset is imbalanced toward positive reviews, we also report ROC-AUC and PR-AUC, which provide a better understanding of model performance across classification thresholds.

We first evaluated baseline text classification models based on TF-IDF features. Logistic Regression achieved a ROC-AUC of approximately 0.853, demonstrating strong performance in deciphering recommended and non-recommended reviews. Naive Bayes produced a slightly lower ROC-AUC of 0.838, but still performed competitively given its simplicity. However, due to the dataset's class imbalance, Naive Bayes seemed to predict "recommended" as the default class, resulting in very high recall for positive reviews (about 0.98), but low recall for negative reviews (0.26). Logistic Regression produced more balanced predictions across classes, making it a stronger baseline overall.

To capture more complex relationships between features, we trained a LightGBM model using three feature configurations: text-only features, metadata-only features, and a combined set of text and metadata features. The text-only LightGBM model resulted in a ROC-AUC of 0.831, while the metadata only model achieved a ROC-AUC of 0.714. This suggests that review text provides the strongest signal for predicting recommendation behavior. When combining both feature sets, the model had the best performance with ROC-AUC of 0.87 and PR-AUC of 0.961, showing that behavior metadata provides complementary predictive information.

The confusion matrix of our best model reveals that the combined text and metadata approach substantially improved detection of negative reviews compared to simpler models.

Recall for negative reviews improved from roughly 50% to 66%, indicating that metadata features helped identify cases where review text alone might appear positive, but user behavior suggests otherwise. Feature importance analysis supports this finding, with variables such as author playtime, review count, and playtime at review time ranking among the most influential predictors.

To verify that our findings generalize beyond a single train–test split, we performed 3-fold stratified cross-validation. The metadata-only model achieved a ROC-AUC of 0.708, the text-only model achieved 0.813, and the combined model achieved 0.860. The combined model again achieved the strongest performance, with consistent scores across folds, suggesting that the model generalizes well to unseen data.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Model	Features	Accuracy	F1	Precision	Recall	ROC-AUC	PR-AUC
Naive Bayes	TF-IDF	0.84	0.90	0.98	0.90	0.838	-
Logistic Regression	TF-IDF	0.81	0.88	0.86	0.88	0.853	-
LightGBM	Text	0.85	0.91	0.88	0.93	0.831	0.94
LightGBM	Metadata	0.74	0.84	0.86	0.81	0.71	0.90
LightGBM	Text+Metadata	0.83	0.89	0.91	0.88	0.870	0.96

Table 1: Model Performance Metrics

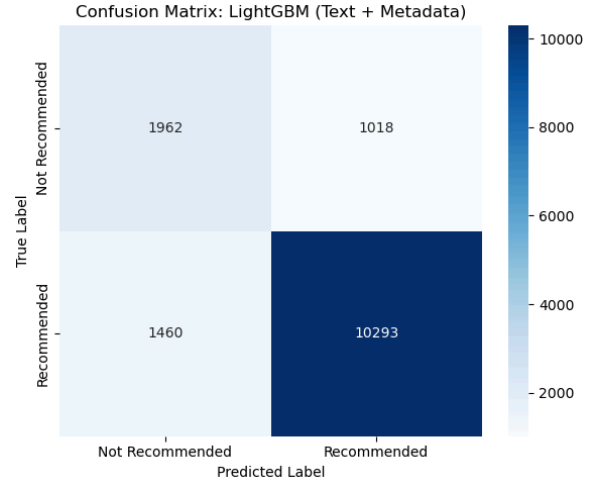


Fig 2. Confusion matrix for the best-performing LightGBM model using both text and metadata features.

B. Helpfulness Prediction

Because review helpfulness prediction is formulated as a regression task, we evaluate model performance using RMSE, MAE, and R^2 . RMSE measures the square root of the average squared prediction error and penalizes large deviations. MAE measures the average absolute difference between predicted and true values. We primarily focus on R^2 , which measures the proportion of variance explained by the model. In this metric, $R^2 = 0$ corresponds to predicting the mean value, while negative values indicate performance worse than the mean baseline.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Below table summarizes the performance of different models for review helpfulness prediction. Using structured features alone, Ridge Regression achieves limited predictive power ($R^2 \approx 0.026$), indicating that linear relationships between reviewer behavior and helpfulness are weak. Decision Tree Regression improves performance slightly ($R^2 \approx 0.046$) by capturing nonlinear interactions between features such as playtime and reviewer history. Among the structured-feature models, Gradient Boosting Regression performs best with $R^2 \approx 0.057$. The ensemble of trees allows the model to capture more complex behavioral patterns in the data. When text features are added, the results differ significantly depending on the representation. TF-IDF + Ridge performs poorly with negative R^2 (≈ -0.004), indicating that simple bag-of-words features do not provide useful signals for predicting review helpfulness. In contrast, BERT embeddings combined with Gradient Boosting achieve the best overall performance ($R^2 \approx 0.067$), suggesting that contextual semantic

representations provide a small but consistent improvement over structured features alone.

Across all five models, R^2 remains below 0.07, meaning that even the best model explains only about 6–7% of the variance in helpfulness scores. This suggests that review helpfulness on Steam is largely driven by platform-level dynamics rather than review content or reviewer metadata.

Ranking	Model	RMSE	MAE	R2
1	BERT + structured + GBR	0.258587	0.240974	0.067174
2	Gradient Boosting	0.251353	0.232497	0.057722
3	Decision Tree	0.252816	0.234982	0.046721
4	Ridge (structured)	0.255515	0.242635	0.026259
5	TF-IDF + structured + Ridge	0.268331	0.243356	-0.004451

Table. 2

BERT + structured + GBR achieves the best R^2 (0.067), while TF-IDF + Ridge falls below the mean baseline ($R^2 = -0.004$).

C. Personalized Game Recommendation

We evaluate recommendation quality using Hit Rate at 10 (HR@10) and Recall@10. HR@10 measures the proportion of test users for whom at least one held-out game appears in the top-10 recommendation list;

$$\text{HR@10} = \frac{1}{|U|} \sum_{u \in U} \mathbf{1}[\mathcal{R}_u \cap \mathcal{T}_u \neq \emptyset]$$

where $\mathcal{R}_u$ is the set of top-10 recommended items for user u and $\mathcal{T}_u$ is the held-out test set. Recall@10 measures the mean fraction of held-out items recovered:

$$\text{Recall@10} = \frac{1}{|U|} \sum_{u \in U} \frac{|\mathcal{R}_u \cap \mathcal{T}_u|}{|\mathcal{T}_u|}$$

For diversity, we additionally report: Catalog Coverage (fraction of unique games recommended across all users), Shannon Genre Entropy (mean per-list entropy over genre distributions, measuring within-list variety), and Mean Popularity Percentile (average popularity rank of recommended games, where lower values indicate more niche recommendations).

All re-ranking methods operate on top-50 ALS candidate lists. The primary hyperparameter for each method is the interpolation weight w , which controls the trade-off between the original ALS score and the re-ranking signal. For Sentiment Boost, $w \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on a held-out validation set; $w=0.5$ maximized HR@10, with performance declining at $w>0.5$ as over-weighting sentiment alignment disrupted the ALS ranking. For BERT re-ranking, the same sweep was conducted; $w=0.3$ was selected, as HR@10 degraded sharply beyond this value, suggesting the semantic signal is informative but noisy relative to the collaborative signal. For Genre+Sentiment diversity re-ranking, the genre penalty weight λ was swept over $\{0.1, 0.3, 0.5\}$; $\lambda = 0.3$ balanced HR@10 retention against catalog coverage and genre entropy gains. No cross-validation was performed for re-ranking weights; a single 80/10/10 train/validation/test split was used consistently across all methods.

To incorporate sentiment information into the recommendation process, VADER compound sentiment scores were computed for each review and aggregated to produce user sentiment profiles: where $\hat{r}_{ui}$ is the ALS predicted score and $\text{sim}(\cdot)$ is cosine similarity between sentiment profile vectors. At $w=0.5$, Sentiment Boost achieved HR@10 = 0.1165 and Recall@10 = 0.0465, a marginal +0.91% improvement over the ALS baseline (HR@10 = 0.1154). This modest gain suggests that sentiment alignment provides a weak but positive signal for re-ranking when not over-weighted.

$$\text{score}(u, i) = (1 - w) \cdot \hat{r}_{ui} + w \cdot \text{sim}(\bar{\mathbf{s}}_u, \bar{\mathbf{s}}_i)$$

User and game semantic profiles were generated by averaging paraphrase-MiniLM-L3-v2 embeddings (384-dimensional) across all associated reviews, resulting in 18,752 user profiles and 15,065 game profiles. PCA was applied to reduce the dimensionality to 50, retaining 74.7% of the variance, to enhance computational efficiency. The re-ranking score follows the same interpolation as above, with $\text{sim}(\cdot)$ as cosine similarity between reduced BERT embeddings. At $w=0.3$, BERT re-ranking achieved HR@10 = 0.1162 and Recall@10 = 0.0463 (+0.69% over baseline). The marginal improvement observed relative to sentiment re-ranking aligns with the use of MiniLM-L3-v2 as a frozen feature extractor rather than being fine-tuned on Steam review language, thereby limiting its domain specificity.

To address the accuracy-diversity trade-off, we implemented a diversity-aware re-ranking step that penalizes genre redundancy within the top-10 list using a Maximal Marginal Relevance (MMR)-style objective. At each ranking position, the next item is selected by

$$i^* = \arg \max_{i \notin \mathcal{R}_u} [(1 - \lambda) \cdot \text{score}(u, i) - \lambda \cdot \text{genre_overlap}(i, \mathcal{R}_u)]$$

where $\text{genre_overlap}(i, Ru)$ is the Jaccard similarity between item i 's genre set and the genres already present in the current list Ru , and $\lambda=0.3$.

Table 4. reports diversity results. Genre+Sentiment achieves $\text{HR@10} = 0.1182$ and $\text{Recall@10} = 0.0552$, with Catalog Coverage improving from 2.49% (baseline) to 3.69% and Shannon Genre Entropy improving from 3.077 to 3.080. For reference, the Popularity Baseline, which recommends the globally top-10 most interacted games to all users, achieves $\text{HR@10} = 0.2197$ but collapses Catalog Coverage to 0.06%, recommending only a handful of unique titles across the entire user base, illustrating the extreme cost of a popularity only strategy. The re-ranking results reveal two key insights. Sentiment and semantic signals offer only slight accuracy gains when applied post-hoc to ALS candidates, indicating they are more effective when integrated during training (as in BM25+ALS). The boost from the voted-up ratio was tested but excluded after a grid search showed it consistently lowered HR@10 , likely adding a popularity bias that inflated scores for widely played games regardless of user preferences. Second, Genre+Sentiment diversity re-ranking improves catalog coverage by 48% with only a 0.86% hit rate cost, making it ideal for practical use where diversity and accuracy both matter.

Method	HR@10	Recall@10	vs.Baseline
Baseline CF	0.1154	0.046	n/a
ALS (pos-only)	0.1780	0.0809	+54.21%
BM25+ALS	0.1918	0.0878	+66.22%
Sentiment Boost (w=0.5)	0.1165	0.0465	+0.91%
BERT Re-ranking (w=0.3)	0.1162	0.0463	+0.69%

Table 3. Accuracy of recommendation methods (full pipeline): Accuracy of recommendation methods across the full pipeline. BM25+ALS achieves the strongest HR@10 (+66.22% over baseline), followed by ALS (pos-only, +54.21%). Re-ranking methods (Sentiment Boost, BERT) show marginal gains over baseline CF

Method	HR@10	Recall@10	Coverage	Genre Entropy
Baseline CF	0.1096	0.0437	2.49%	3.077
Sentiment Boosted	0.1116	0.0444	2.62%	3.079
Genre+Sentiment	0.1182	0.0552	3.69%	3.080
Popularity Baseline	0.2197	0.1190	0.06%	2.632

Table 4. Diversity analysis across re-ranking methods: Diversity analysis across re-ranking methods. Results are evaluated on the diversity analysis user sample which differs from Table 3. evaluation set; HR@10 values are not directly comparable across tables

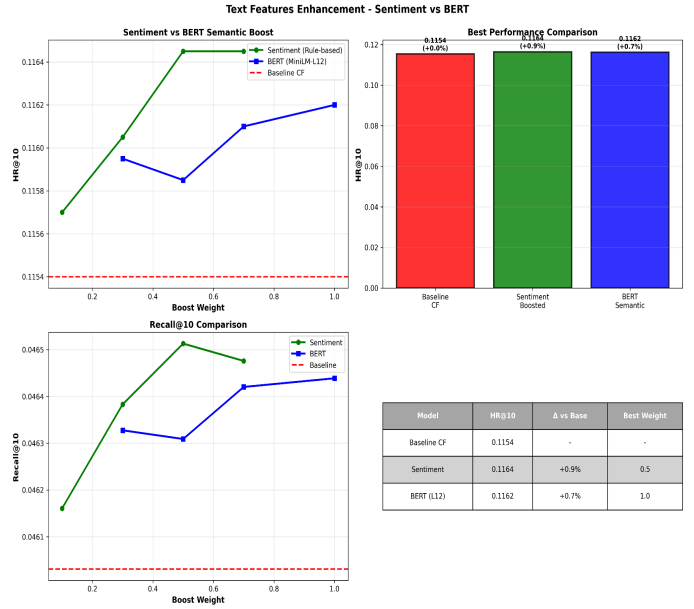


Fig. 3. HR@10 as a function of re-ranking boost weight for Sentiment (VADER) and BERT (MiniLM-L3-v2) re-ranking methods. The dashed red line indicates the baseline CF performance ($\text{HR@10} = 0.1154$). Sentiment peaks at $w = 0.5$ and BERT peaks at $w = 0.3$, with both methods degrading at higher weights.

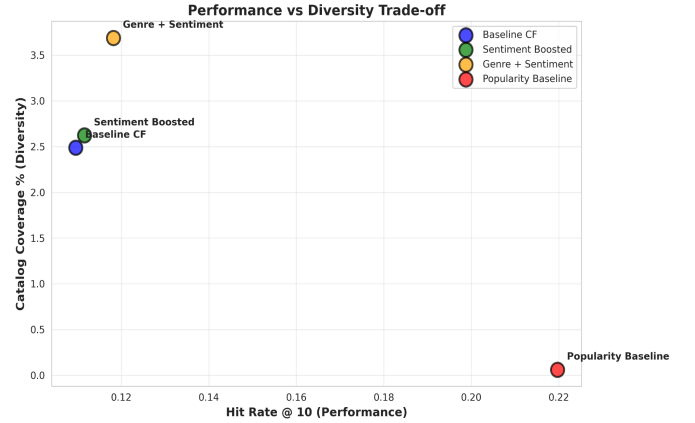


Fig. 4. Performance vs. diversity trade-off across recommendation methods. HR@10 (x-axis) is plotted against Catalog Coverage % (y-axis). The Popularity Baseline achieves the highest HR@10 (0.2197) at the cost of near-zero catalog coverage (0.06%), while Genre+Sentiment achieves the best trade-off among re-ranking methods ($\text{HR@10} = 0.1182$, Coverage = 3.69%).

D. Error Analysis

Recommendation Classification: despite strong overall performance, several limitations remain. Mixed-sentiment reviews can be difficult for TF-IDF-based models because word frequency does not capture the overall narrative structure of the review. A review may criticize the game, but still ultimately recommend it. Short or non-English reviews can also reduce performance, as they act as limited or noisy textual signals. Overall, the experiments show that review text is the strongest signal for predicting whether a user recommends a game, while behavioral metadata provides useful information that improves detection of negative reviews. Future work could explore fine-tuning contextual language models such as

BERT on the Steam dataset to better capture sarcasm, contextual meaning, and mixed sentiment.

Helpfulness Prediction: To further understand model behavior, we analyze the residual distributions. The residual plots for both TF-IDF + Ridge and BERT + GBR reveal a clear bimodal pattern, with one peak around -0.3 and another around $+0.2$. This pattern directly reflects the zero-inflation in the ‘weighted_vote_score’ variable: more than 50% of reviews receive a score of 0. For these reviews, the model often predicts small positive values, creating the right-side residual peak. Conversely, reviews with high helpfulness scores tend to be underestimated by the model, producing the left-side peak. This structural bias appears consistently across all models and cannot be eliminated simply by increasing model complexity. Several missing contextual signals may explain this limitation. First, many reviews receive zero helpfulness votes not because they are unhelpful, but because they were never seen by enough users to receive votes. Second, review visibility and timing strongly influence voting behavior: reviews posted close to a game’s release receive far more exposure. Third, Steam’s ‘weighted_vote_score’ incorporates ‘internal trust signals’ that are not observable in our dataset. Finally, ‘game popularity’ plays an important role, as reviews for popular titles attract significantly more readers and votes.

Personalized Game Recommendation: Re-Ranking Signal dilution is the most common failure mode in both sentiment analysis and BERT re-ranking. The ALS model uses interaction frequency as a dense, reliable signal to capture implicit positive preferences. Post-hoc re-ranking adds a weaker secondary signal that, if overemphasized, can override correct ALS recommendations with misaligned items. This is evident in both methods: Sentiment Boost reaches its peak at $w = 0.5$. BERT re-ranking peaks at $w=0.3$, with performance declining steadily beyond this point.

Explicit Dislike: ALS is a strong collaborative filtering model but it has a key limitation in implicit-feedback settings. It is not able to directly learn dislike signals. In our dataset, interactions were represented by positive engagement, the absence of an interaction doesn’t necessarily indicate that a user dislikes a game. It could just mean that the user has not encountered it, as a result of this ALS models primarily learn the patterns of positive preference rather than modeling negative feedback. This limitation can reduce precision in situations where users are not fans of a specific genre or title; however, focusing on positive interactions can be beneficial in settings like game recommendations. By emphasizing items similar to those that a user has enjoyed in the past, the model is able to recommend a wider set of relevant games that the user might enjoy. We experimented with including negative interactions into the model but it did not improve performance.

Limitations of Frozen BERT Embeddings: MiniLM-L3-v2 was used as a frozen feature extractor without fine-tuning on Steam review language. The embeddings capture general semantic similarities but miss domain-specific features like gaming vocabulary, genre conventions, and platform-specific sentiments. This limitation weakens the BERT re-ranking

signal and explains the slight HR@10 improvement despite ample review text.

VADER Sentiment as a Coarse Signal: Compound scores consolidate review sentiment into a single scalar value, thereby losing nuances such as mixed sentiments (e.g., "great story, terrible performance") or sentiment targets specific to the game. Users whose preferences are shaped by specific game attributes rather than overall sentiment are consequently insufficiently served by the sentiment boost re-ranking, which explains its limited effectiveness.

VI. CONCLUSION

This project presents a multi-task, multi-modal analysis of Steam user reviews through three complementary tasks: recommendation classification, review helpfulness prediction, and personalized game recommendation. Across all tasks, our results show that combining textual information with behavioral metadata generally provides stronger performance than relying on a single modality alone. For recommendation classification, LightGBM achieved the strongest results by capturing nonlinear interactions between TF-IDF features and user/game metadata, substantially improving recall for negative reviews compared with simpler text-only baselines. For helpfulness prediction, Gradient Boosting with BERT embeddings performed best, but the overall explanatory power remained limited, suggesting that review helpfulness is driven largely by external platform-level factors such as visibility, timing, and popularity rather than review content alone. For personalized recommendation, BM25-weighted ALS produced the highest ranking accuracy, while Genre+Sentiment re-ranking offered the best balance between recommendation quality and diversity by improving catalog coverage without a major loss in hit rate. Overall, our findings highlight that multimodal signals are effective for modeling user behavior on Steam, but they also reveal important limitations caused by missing contextual information and popularity bias. Together, these findings demonstrate how combining text, metadata, and collaborative signals can support more accurate and more diverse gaming platform recommendations. Future work should explore cross-task integration and domain-specific fine-tuning of semantic models. For example, incorporating helpfulness scores as interaction weights in collaborative filtering, or fine-tuning BERT on Steam review language, could further close the performance gap.

VII. CONTRIBUTION

Bokyoung Son: Led project coordination, milestone planning, and inter-team communication throughout the duration of the Capstone Project. Responsible for developing and drafting the outlines of the final report, ensuring consistency across sections and adherence to IEEE formatting standards. Co-contributed to Task 3 (Personalized Game Recommendation) alongside Brandon Rocchio, with primary responsibility for the sentiment-based re-ranking pipeline (VADER Sentiment Boost), BERT semantic re-ranking (paraphrase-MiniLM-L3-v2), and genre- and sentiment-aware diversity re-ranking. Conducted a comprehensive diversity analysis comparing catalog coverage, Shannon genre entropy, intra-list diversity, and popularity bias across all recommendation methods. Additionally contributed to the data

preprocessing pipeline and semantic feature engineering, including the generation of BERT embeddings and the construction of VADER sentiment profiles.

Brandon Rocchio: Responsible for training the baseline recommendation model and contributing to task 3 (Personalized Game Recommendation) alongside Bokyoung Son with a focus on collaborative filtering using the alternating least squares algorithm. Contributed to the initial discovery and preprocessing of our dataset to meet the space requirements in our DataHub server. Supported Bokyoung in leading group coordination and timing for group meetings throughout the project preparation.

Hangyi Zhao: Responsible for data preprocessing and dataset preparation, including cleaning the raw Steam review data, restructuring the dataset, and converting it into Parquet format for efficient processing. Conducted exploratory data analysis (EDA). Led Task 2: Review Helpfulness Prediction, designing and implementing several modeling pipelines. This included developing baseline models such as Ridge Regression and Decision Tree, as well as a Gradient Boosting regression model to capture nonlinear relationships in structured features. Text representations were incorporated using TF-IDF features and BERT embeddings to evaluate the impact of semantic information from review text. In addition, supported team coordination and inter-team communication throughout the duration of the Capstone project.

Paige Saengsouvanna: Conducted EDA. Responsible for Task 1: Recommendation Review. Designed the pipeline for the task, starting with our baseline models, advanced LightGBM models, evaluation, cross validation, and feature extraction.

VIII. REFLECTIONS

Reflecting on the Capstone project, several decisions clearly offer room for improvement. Limiting the project to two tasks instead of three would have allowed for deeper exploration and more rigorous experimentation in each. Pursuing recommendation classification, helpfulness prediction, and personalized recommendation simultaneously created time constraints that limited hyperparameter tuning, cross-task integration, and model iteration.

Second, the BERT model used for semantic re-ranking was a frozen feature extractor, not fine-tuned on Steam review language. Fine-tuning Paraphrase-MiniLM-L3-v2 on Steam reviews would likely yield more domain-specific embeddings and a stronger re-ranking signal, but this wasn't feasible due to the DSMLP cluster's computational limits.

Third, genre-specific models, proposed in the project abstract, were not implemented due to an 18% missing genre rate after the AppID join and cold-start issues with less-reviewed games. Spending more time on cleaning and imputing genre data early in the project could have made this goal attainable.

Fourth, this work independently developed and evaluated sentiment-based and BERT-based re-ranking approaches. In

hindsight, combining VADER sentiment alignment, BERT cosine similarity, and genre affinity into a weighted hybrid re-ranking model could have outperformed any single signal. Systematically exploring hybrid formulations with learned combination weights instead of manual boost parameters is a clear direction for future research.

Finally, the three tasks were developed mainly in parallel as separate pipelines, not as parts of an integrated system. Designing tasks to share representations from the start—such as using helpfulness scores from Task 2 to weight interactions in Task 3—would have made the contribution more coherent and innovative.

IX. REFERENCES

- [1] T. Baltrušaitis, C. Ahuja, and L. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, 2019.
- [2] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers," in *Proc. NAACL*, 2019.
- [3] J. Du et al., "Feature selection for helpfulness prediction of online product reviews," *PLoS ONE*, vol. 14, no. 12, 2019.
- [4] S. Hassan et al., "Studying the dialogue between users and developers of free apps," *Empirical Softw. Eng.*, vol. 23, no. 3, 2018.
- [5] C. J. Hutto and E. Gilbert, "VADER: A parsimonious rule-based model for sentiment analysis," in *Proc. ICWSM*, 2014.
- [6] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, 2009.
- [7] K. Kuan et al., "What makes a review voted?," *J. Assoc. Inf. Syst.*, vol. 16, no. 1, 2015.
- [8] D. Lin et al., "An empirical study of game reviews on the Steam platform," *Empirical Softw. Eng.*, vol. 24, no. 1, 2019.
- [9] S. McIlroy et al., "Is it worth responding to reviews?," *IEEE Softw.*, vol. 34, no. 3, 2017.
- [10] S. M. Mudambi and D. Schuff, "What makes a helpful online review?," *MIS Q.*, vol. 34, no. 1, 2010.
- [11] M. J. Pazzani and D. Billsus, "Content-based recommendation systems," in *The Adaptive Web*, Springer, 2007.
- [12] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Found. Trends Inf. Retr.*, vol. 3, no. 4, 2009.
- [13] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, 2017.
- [14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019.
- [15] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, 2011.

[16] B. Frederickson, "implicit: Fast Python Collaborative Filtering for Implicit Datasets," GitHub, 2021. [Online]. Available: <https://github.com/benfred/implicit>

potential improvement for capturing domain-specific language.

[17] N. Reimers and I. Gurevych, "Making monolingual sentence embeddings multilingual using knowledge distillation," in Proc. EMNLP, 2020.

[18] Apache Arrow, "PyArrow: Python library for Apache Arrow," Apache Software Foundation, 2016. [Online]. Available: <https://arrow.apache.org>

[19] W. McKinney, "Data structures for statistical computing in Python," in Proc. SciPy, 2010.

[20] C. R. Harris et al., "Array programming with NumPy," Nature, vol. 585, 2020.

[21] P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," Nature Methods, vol. 17, 2020.

[22] L. Eberhard et al., "Investigating Helpfulness of Video Game Reviews on the Steam Platform," in Proc. 2018 Fifth Int. Conf. on Social Networks Analysis, Management and Security, 2018, pp. 43–50.

[23] Z. Wang et al., "Explaining and Predicting Helpfulness and Funniness of Online Reviews on the Steam Platform," J. Global Inf. Manage., vol. 29, no. 6, 2021.

[24] M. K. Baowaly et al., "Predicting the helpfulness of game reviews: A case study on the Steam store," J. Intell. Fuzzy Syst., vol. 36, no. 5, pp. 5087–5097, 2019.

[25] K. Poc, "Steam Reviews Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/kieranpoc/steam-reviews>

[26] F. Kong, "Steam Games Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/fronkongames/steam-games>

X. REPLY TO PEER REVIEWS

Based on the suggestions, we see how there could be several improvements to clarify and strengthen the project. We discussed how our project objectives - recommendation prediction, helpfulness prediction, and game prediction - relate to one another and how signals from one task could inform another. For instance, helpfulness scores could potentially be used to weight more informative reviews in the recommender system, while recommendation predictions could offer additional signals when ranking games. We also clarified the limitations regarding short and non-English reviews and noted that future improvements could incorporate language detection or multilingual models to better preserve sentiment information. We also addressed modeling considerations and concerns in the review. In the helpfulness prediction task, we acknowledged the low R^2 value and discussed how reframing the problem as a binary classification task (helpful vs. not helpful) may produce stronger predictive results. Additionally, we clarified that BERT embeddings were used as contextual features rather than fine-tuning the model directly. Given more time, fine-tuning BERT on the Steam dataset is a